

Initial implementation and testing

Urban sound event detection on RK3588 using pretrained YAMNet

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GitHub Project Site

<https://kolentossa.github.io/elec5305-project-550374130/>

Project description

My project tests prerecorded WAV files on RK3588 using pretrained YAMNet. It focuses on converting class scores into sound labels and nominal event intervals, and comparing simple temporal decisions for recorded audio. The initial classes are dog barking, sirens and car horns.

Implementation and preliminary results

Six controlled scenes totalling 56 seconds were tested on the RK3588 NPU (LubanCat-5 V2) and Windows CPU. Each backend retained 119 patches of 521 scores. The NPU used a converted FP16 model with RKNN Runtime 2.3.2. Independent thresholding at 0.2 was compared with causal exponential smoothing at alpha 0.5; these are fixed, untuned parameters.

Both methods on both backends gave recording-level micro precision 100.0% and recall 85.7%, with TP 6, FP 0 and FN 1 across 18 scene-label decisions. A label is expected when its source file is inserted and predicted when at least one interval is produced anywhere in the scene.

The missed label was the horn in the mixed scene: its NPU score peaked at 0.078, compared with 0.876 alone. Source amplitude also changed, so the cause is unresolved. Smoothing extended nominal interval endings by 0.48 seconds without changing interval counts. Six unit checks passed on Windows and RK3588.

Limitations and next steps

The pilot reuses only three source recordings. Its metrics assess label presence; audible event boundaries are unannotated, so event boundary accuracy is unmeasured. Next, I will add annotated URBAN-SED tests, select parameters on the validation set, compare class-specific thresholds and hysteresis, and test further recordings on RK3588. Code, results and reproduction instructions are available through the project site.